

Test Plan Documentation

Subgroup 3: Devices

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Revision History

Date	Version	Description	Author
08/May/26	1.0	Updated file to contain Table of content and figures.	Mustafa Al-Bayati, Sergej Macut, Tony Nguyen, Ryad Hazin
08/May/26	1.1	Initial draft of the test plan, defining unit testing and documentation scope for the ARDUINO_REFACTOR project.	Mustafa Al-Bayati, Sergej Macut, Tony Nguyen, Ryad Hazin
08/May/26	1.2	Added PlatformIO native unit testing setup and Unity framework configuration.	Sergej Macut Ryad Hazin Tony Nguyen
08/May/26	1.3	Added device unit tests for BuzzerDevice, Fan, DoorDevice, and WindowDevice.	Mustafa Al-Bayati

Test Plan

Task Delegation

Here is a balanced breakdown of the unit testing and documentation workload, divided by the folders and files shown in your `src` directory:

- **Ryad** will be responsible for the core logic, configuration, and system safety. This includes writing unit tests and documentation for the `core/` folder (`CommandHandler.cpp`, `CommandHandler.h`, `Observer.h`, `Subject.h`), the `config/` folder (`config.cpp`, `config.h`), and the `safety/` folder (`GasSafety.cpp`, `GasSafety.h`) Figure[3].
- **Mustafa** will handle the hardware device representations. Due to the size of the folder, Mustafa will focus entirely on the `devices/` folder, covering tests and documentation for `BuzzerDevice`, `Devices` (base/manager), `DoorDevice`, `Fan`, and `WindowDevice` Figure[4] and [5].
- **Sergej** will take charge of the system inputs and safety mechanisms. This involves unit testing and documenting the `sensors/` folder (`sensors.cpp`, `sensors.h`, `SteamSensor.cpp`, `SteamSensor.h`), the `music/` folder (`MusicEngine.cpp`, `MusicEngine.h`), and the entry point file (`main.cpp`) Figure[2].
- **Tony** will manage the outputs, media, and the main execution loop. Tony will write tests and documentation for the `display/` folder (`LCDManager.cpp`, `LCDManager.h`), and the `gateway.js` file.

Figures

```
===== SUMMARY =====
Environment  Test      Status  Duration
-----
uno          test_music SKIPPED
uno          test_sensors SKIPPED
native       test_music PASSED    00:00:01.383
native       test_sensors PASSED    00:00:01.426
===== 8 test cases: 8 succeeded in 00:00:02.809 =====
```

Figure 2: Music and Sensors tests

```
===== SUMMARY =====
Environment  Test      Status  Duration
-----
uno          test_config PASSED    00:00:00.332
uno          test_core PASSED    00:00:10.646
uno          test_safety PASSED    00:00:09.166
===== 10 test cases: 10 succeeded in 00:00:29.144 =====
```

Figure 3: Config, Core and Safety tests

```
musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio test -e native
Verbosity level can be increased via '-v, -vv, or -vvv' option
Collected 1 tests

Processing test_devices in native environment
-----
Building...
Testing...
test\test_devices\test_main.cpp:18: test_buzzer [PASSED]
----- native:test_devices [PASSED] Took 1.06 seconds -----

===== SUMMARY =====
Environment  Test      Status  Duration
-----
native       test_devices PASSED    00:00:01.062
===== 1 test cases: 1 succeeded in 00:00:01.062 =====

musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
```

Figure 4: Devices testing

```
musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio run -e native -t clean
native SUCCESS 00:00:00.555
===== 1 succeeded in 00:00:00.555 =====

musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio test -e native
Verbosity level can be increased via '-v, -vv, or -vvv' option
Collected 1 tests

Processing test_devices in native environment
-----
Building...
Testing...
test\test_devices\test_main.cpp:77: test_buzzer_turns_on [PASSED]
test\test_devices\test_main.cpp:78: test_buzzer_turns_off [PASSED]
test\test_devices\test_main.cpp:80: test_fan_turns_correct_direction [PASSED]
test\test_devices\test_main.cpp:82: test_door_closes [PASSED]
test\test_devices\test_main.cpp:84: test_window_closes [PASSED]
----- native:test_devices [PASSED] Took 1.27 seconds -----

===== SUMMARY =====
Environment  Test      Status  Duration
-----
native       test_devices PASSED    00:00:01.273
===== 5 test cases: 5 succeeded in 00:00:01.273 =====
```

Figure 5: Buzzer, Door, Fan and Window

```

esting -vvv
Flash: [=          ] 11.7% (used 3774 bytes from 32256 bytes)
.pio\build\uno\firmware.elf :

section                size      addr
.data                  370      8388864
.text                  3404         0
.bss                   223      8389234
.comment               17         0
.note.gnu.avr.deviceinfo 64         0
.debug_aranges         256         0
.debug_info            3020         0
.debug_abbrev          1602         0
.debug_line            1290         0
.debug_str             520         0
Total                  10766

----- uno:* [PASSED] Took 1.38 seconds -----

===== SUMMARY =====
Environment   Test    Status   Duration
-----
uno           *      SKIPPED  00:00:01.385
native       *      SKIPPED
===== 0 test cases: 0 succeeded in 00:00:01.385 =====

```

```

PASS test/gateway/gateway.test.js
Gateway.js - Full Logic Suite
  Basic Helpers
    ✓ normalize: should trim and lowercase (4 ms)
    ✓ normalizeBoolean: should recognize 'on'/'off' strings
    ✓ clampInt: should respect boundaries
  Music Logic
    ✓ parseNotes: should parse semicolon separated string (2 ms)
    ✓ serializeNotes: should format for Arduino transmission (1 ms)
    ✓ getMusicState: should extract play and notes from complex data (1 ms)
  State Management
    ✓ getFirestoreCommandState: should map device states correctly (2 ms)
    ✓ hasChanges: should detect differences between objects (2 ms)
Test Suites: 1 passed, 1 total
Tests:      8 passed, 8 total
Snapshots:  0 total
Time:       0.59 s, estimated 1 s
Ran all test suites.

```

Figure 7. Gateway test